LIMITS AT INFINITY

Limits at infinity vs infinite limits

Both use the infinity symbol, but in opposite roles.

Limit at infinity

NOTATION

$\lim_{x\to\infty} f(x)$ or $\lim_{x\to-\infty} f(x)$

WHAT ESCAPES

The input: x grows without bound while we track where $f(x)$ goes

WHERE THE SYMBOL SITS

Under the arrow

EXAMPLE

$\lim_{x\to\infty} \frac{1}{x} = 0$

Infinite limit

NOTATION

$\lim_{x\to a} f(x) = \infty$ or $= -\infty$

WHAT ESCAPES

The output: x approaches a finite a while $f(x)$ grows without bound

WHERE THE SYMBOL SITS

After the equals sign

EXAMPLE

$\lim_{x\to 0} \frac{1}{x^2} = \infty$

≠ **OPPOSITE ROLES**

What tells them apart is which side escapes: the input or the output.

The two can combine: in $\lim_{x\to\infty} x^2 = \infty$ the input and the output both escape, a limit at infinity whose value is infinite.